ASTIBE, JULIANA EUNICE A.

BSIT 2-1

ACTIVITY #1 - Variables

COMP 009 - Object Oriented Programming

PART A - VARIABLE IDENTIFICATION TABLE

Variable Name	Data Type	Variable Type	Stored In
name	String	Instance	Heap
grade	int	Instance	Heap
s1	Student	Local	Stack
s2	Student	Local	Stack
s3	Student	Local	Stack

PART B - Explanation. Please see example below.

Statement	What Happens?	Memory Impact
Student s1 = new Student("John", 85);	A new Student object is created & s1 points to it.	Stack: s1 holds memory address. Heap: the object name "John", grade = 85.
Student s2 = new Student ("Maria", 90);	A second Student object is created & s2 points to it.	Stack: s2 holds different memory. Heap: object has name = "Maria", grade = 90.
Student s3 = s2;	Here, s3 points to the same as what s2 points to.	Stack: s3 points to the same memory address as s2. Heap: name = "Maria", grade = 90.
s1.grade = 95;	The grade pointed to by s1 is updated to 95.	Heap: Object referenced by s1 changes grade from 85 to 95.
s3.name = "Ana";	The name pointed to by s3 is updated to "Ana".	Heap: Object referenced by s3 (and s2) changes "Maria" to "Ana".

PART C - OUTPUT

John 95

Ana 90

Ana 90